

Project Title: Battery Level Indicator

Project Overview:

The **Battery Level Indicator** project is an electronic circuit designed to monitor the voltage level of a battery and provide a visual indication of its charge status using an **LM339 comparator IC** and an **LED**. The circuit compares the battery voltage with a preset reference voltage generated by a potentiometer. When the battery voltage reaches or falls below the set threshold, the comparator changes its output state, causing the LED to turn on or off. This simple and low-cost project helps users monitor battery condition and avoid unexpected battery failure.

Objective

The objective of this project is to design a simple and low-cost battery level indicator using an LM339 comparator IC to monitor a 9V battery's voltage. The circuit provides a visual indication through an LED, helping users identify when the battery voltage falls below a preset level and needs replacement or recharging.

Components Used:

1. Breadboard.
2. 9V Battery
3. LM339 Comparator IC
4. LED
5. Potentiometer
6. Resistor
7. Jumper Wires.

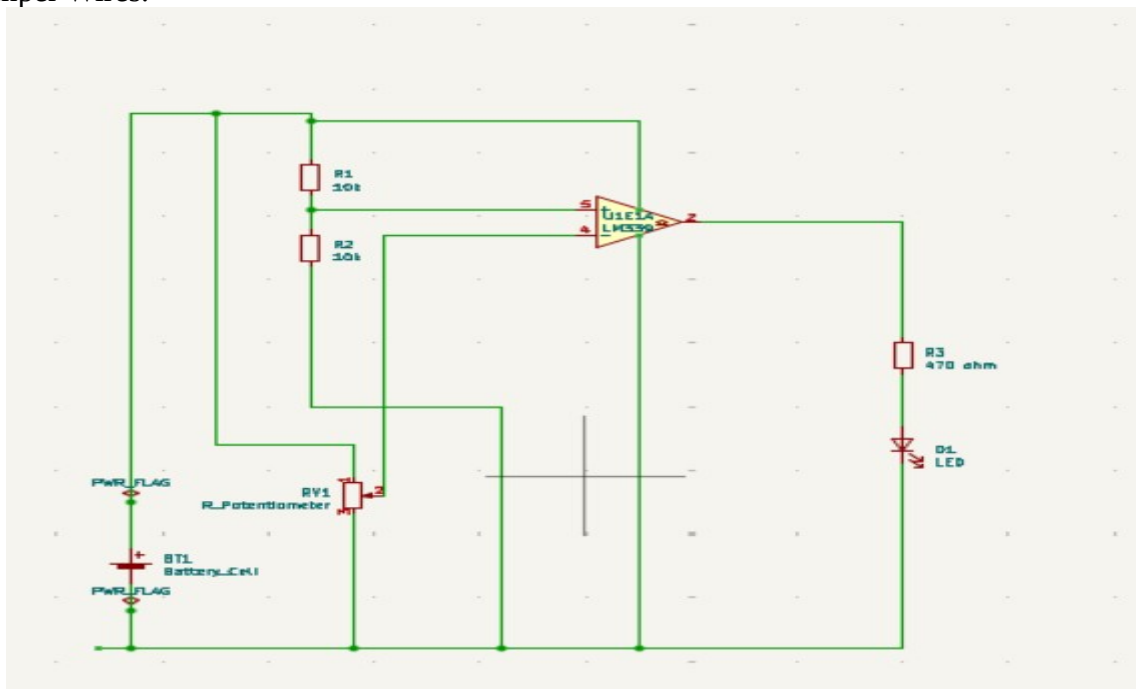


Figure: KiCad Schematic Design of Battery Level Indicator.

Design Tools Used

- KiCad (Schematic Design)
- Tinkercad (Circuit Simulation)
- GitHub (Project Repository)

Circuit Description:

The Battery Level Indicator circuit is built using an LM339 comparator IC, a 9V battery, a 10 k Ω potentiometer, an LED, and resistors. The 9V battery powers the circuit, while the potentiometer is used to set a reference voltage. The LM339 compares the battery voltage with this reference voltage. If the battery voltage is above the preset level, the comparator keeps the LED in one state; when the battery voltage falls below the reference voltage, the comparator changes its output, causing the LED to indicate that the battery is low. A current-limiting resistor protects the LED from excessive current. The entire circuit is assembled on a breadboard using jumper wires, making it easy to build, test, and adjust.

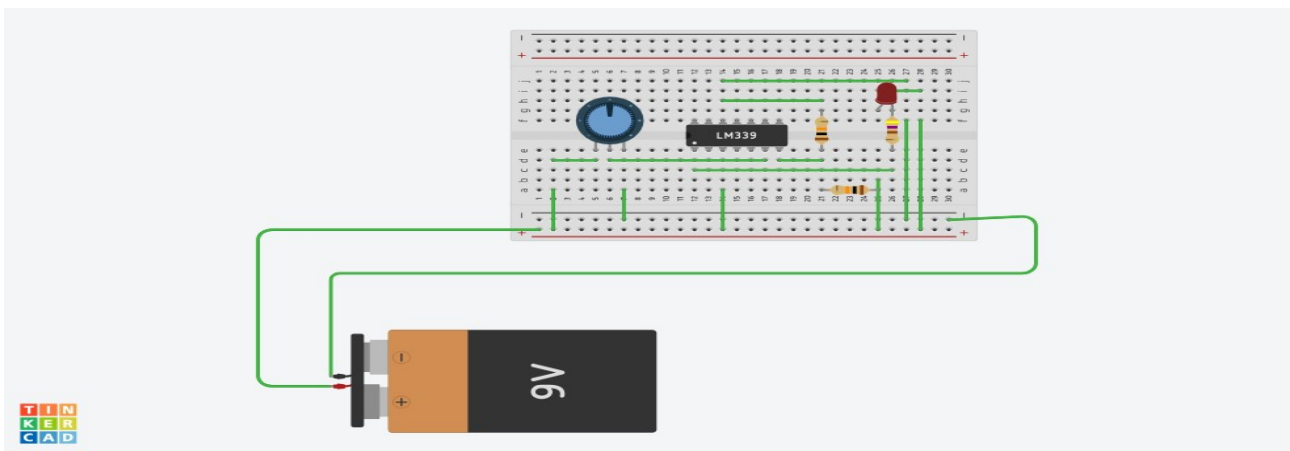


Figure: Tinkercad Simulation of Battery Level Indicator.

Working Principle

1. **Voltage Monitoring:** The 9V battery supplies power, and the battery voltage is applied to one input of the LM339 comparator.
2. **Reference Setting:** The potentiometer provides an adjustable reference voltage to the other input of the comparator.
3. **Voltage Comparison:** The LM339 compares the battery voltage with the preset reference voltage.
4. **Output Indication:** If the battery voltage falls below the reference level, the comparator changes its output and the LED turns ON (or OFF, depending on the circuit), indicating a low battery condition.

Result

The Battery Level Indicator circuit was successfully designed and tested. The LM339 comparator accurately monitored the battery voltage and controlled the LED based on the preset reference voltage. The LED provided a clear visual indication of the battery's condition, demonstrating the circuit's effective operation.

Conclusion

The Battery Level Indicator is a simple, reliable, and low-cost circuit for monitoring battery voltage. It effectively indicates when a battery voltage falls below a preset level, helping users identify when the battery needs recharging or replacement. This project also enhances understanding of comparator circuits, voltage monitoring, and basic electronic design principles.